Reformulating KOTE for Korean Sentiment Classification: A Comparison of Traditional, Multilingual, and Korean-Specific Models

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Abstract

This project studied Korean sentiment classification using the KOTE dataset after simplifying its original fine-grained emotion labels. Because KOTE contains 43 multi-label emotion categories, the labels were consolidated into three broader sentiment classes: positive, negative, and neutral. I compared a TF-IDF baseline with multilingual and Korean-specific transformer models, including multilingual BERT, XLM-RoBERTa, and KLUE-RoBERTa. The results show that transformer-based models clearly outperform the traditional baseline, and KLUE-RoBERTa achieves the best overall performance. However, all models struggled with the neutral class. Error analysis suggests that this was caused by both class imbalance and the fact that the neutral category became semantically mixed after label consolidation. These findings show that while simplification makes the task more practical, label design still has a strong effect on model performance.

1.Introduction

Sentiment analysis is widely used in many fields to understand people's opinions and emotions through text, such as social media posts, survey responses, and customer reviews. It is important in both industry and academic research because it helps turn unstructured text into useful information about people's attitudes, reactions, and experiences.

For Korean text, sentiment and emotion analysis is more challenging because high-quality Korean resources are more limited than English resources. As a result, Korean text is often analyzed using multilingual models or by translating the text into English and then applying English-based NLP models. These methods are practical, but they may not fully preserve the original meaning of Korean text.

This issue is especially important for emotion-related tasks because emotions are often closely related to culture, context, and subtle linguistic nuance. A sentence that implies a certain emotion in Korean may lose intensity or shift meaning when translated into another language. This raises the question of how Korean emotion data should be modeled when the original labels are very detailed and multi-label. More specifically, it is unclear whether multilingual or translation-based approaches are sufficient for this type of task, or whether Korean-specific pretrained models are better suited after the data is reformulated into broader sentiment categories.

This project uses the KOTE (Korean Online That-gul Emotions) dataset, which contains Korean online comments labeled with 43 fine-grained emotions. Because the original labels are highly detailed and multi-label, this project consolidates them into three broader sentiment categories: positive, negative, and neutral. It then compares a traditional baseline model, multilingual transformer models, and a Korean-specific transformer model to evaluate how well each approach performs on Korean sentiment polarity classification.

2.Background

Traditional text classification methods such as TF-IDF with logistic regression have been used as strong baselines for sentiment analysis. Wang and Manning (2012) showed that simple bag-of-words methods can still perform competitively on many text classification tasks, which makes them useful baselines when evaluating more advanced models. At the same time, pretrained transformer models such as BERT have significantly improved performance on many NLP tasks by using contextualized representations rather than only surface-level word features (Devlin et al., 2019).

For multilingual settings, models such as multilingual BERT and XLM-RoBERTa have been developed to transfer knowledge across languages. Conneau et al. (2020) showed that multilingual pretrained models can achieve strong performance on cross-lingual tasks and that XLM-RoBERTa outperformed multilingual BERT on several benchmark tasks. These models are useful for Korean NLP because they can benefit from large multilingual pretraining corpora, even when Korean-specific resources are relatively limited.

However, emotion analysis is more difficult than broad sentiment classification because emotions often depend heavily on linguistic and cultural nuance. The KOTE dataset was introduced to address this problem for Korean by providing 50,000 Korean online comments labeled with 43 fine-grained emotions and one no-emotion category (Jeon et al., 2024). While KOTE makes fine-grained Korean emotion analysis possible, its large label space and multi-label structure also make the task more complicated. This project builds on that work by simplifying the original label space into broader sentiment categories and comparing how traditional, multilingual, and Korean-specific models perform under that reformulation.

3.Methods

3.1 Dataset and Label Consolidation

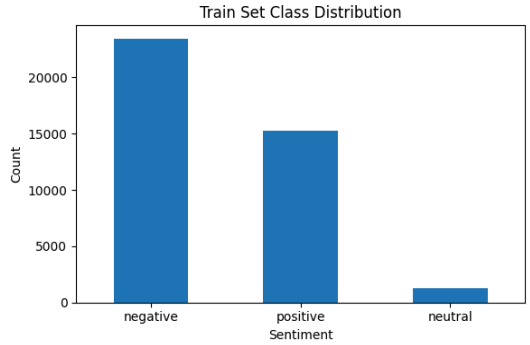
I used the KOTE dataset from Hugging Face, which contains Korean online comments and their emotion labels. The dataset includes 43 fine-grained emotion categories and one no-emotion category. Some examples of the emotion categories include 슬픔 (sadness), 화남/분노 (anger), 불안/걱정 (anxiety/worry), 고마움 (gratitude), 행복 (happiness), 기쁨 (joy), 놀람 (surprise), and 없음 (no emotion). Because each comment can have multiple emotions, the dataset is naturally a multi-label classification task rather than a single-label task. KOTE is appropriate for this project because it was specifically designed for Korean emotion analysis and contains a broader range of emotion labels than earlier Korean resources (Jeon et al., 2024).

Because the original KOTE labels are highly detailed and each text can contain multiple labels, I simplified the label space before training the models. The KOTE paper includes an interpretation of emotion clusters in Appendix A, and I used this as the basis for consolidating the original fine-grained emotion labels into three broader sentiment categories: positive, negative, and neutral. I then applied a score-based rule to assign one final label to each comment. Positive labels were assigned a score of +1, negative labels a score of -1, and neutral labels a score of 0. If the total score was greater than 0, the comment was labeled as positive; if the total score was less than 0, it was labeled as negative; and if the score was 0, it was labeled as neutral.

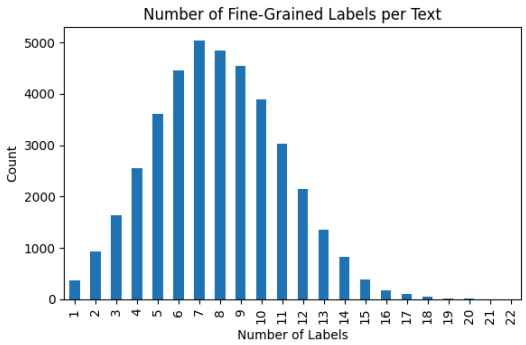
3.2 Exploratory Data Analysis

To understand the structure of the consolidated dataset, I performed exploratory data analysis. This included examining class balance, the number of original labels per comment, and text length distributions across the three sentiment categories. The analysis showed that the original KOTE annotations were strongly multi-label, with the average number of labels per comment being about 8. The consolidated dataset also remained imbalanced, especially for the neutral class. In the training set, the negative class accounted for 58.6% of the data, the positive class accounted for 38.1%, and the neutral class accounted for only 3.3%. Text lengths were broadly similar across classes, although negative comments showed a larger distribution and a longer tail.

(a) Histogram of the class distribution



(b) Histogram of number of labels per text



(c) Distribution of text length

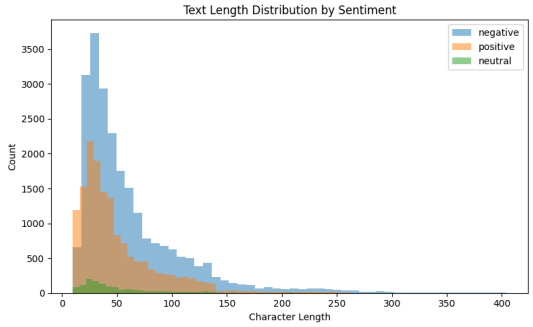


Figure 1. EDA of the KOTE dataset: (a) class distribution across the three sentiment categories, (b) distribution of the number of labels per text in the original dataset, and (c) distribution of text length.

3.3 Models

For the baseline, I used TF-IDF features with multiclass logistic regression, which is a standard baseline for text classification (Wang & Manning, 2012). I then fine-tuned three transformer-based models: multilingual BERT, XLM-RoBERTa, and KLUE-RoBERTa. These models were chosen to compare a traditional baseline, multilingual pretrained models, and a Korean-specific pretrained model. Transformer-based models were selected because pretrained contextual representations have shown strong performance on both cross-lingual and monolingual language understanding tasks (Conneau et al., 2020).

This project fine-tuned pretrained transformer models for a simpler 3-class sentiment classification task after relabeling the data into positive, negative, and neutral categories. I used the Hugging Face `AutoModelForSequenceClassification` implementation, which automatically adds a classification head. The models were trained with a maximum input length of 128, batch size of 16, learning rate of 2e-5, weight decay of 0.01, and 3 training epochs. The best model was selected based on macro F1 score on the validation set.

I also considered an additional translation-based English BERT experiment, but pilot translations generated by a Korean-to-English model showed semantic distortion and were not suitable for reliable emotion classification. Therefore, the main comparison focused on traditional, multilingual, and Korean-specific models applied directly to Korean text.

3.4 Evaluation Metrics

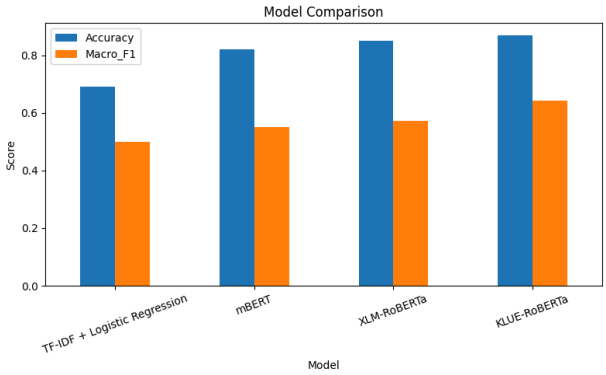
For evaluation, I used accuracy, macro F1 score, classification reports, and confusion matrices. Macro F1 was an important metric because the sentiment classes were imbalanced, and overall accuracy alone would not fully capture performance on the smaller neutral class. I also performed error analysis, especially for the neutral class, to better understand why some sentiment categories were more difficult to classify than others.

4. Results and Discussion

Model	Accuracy	Macro_F1	Negative_F1	Positive_F1	Neutral_F1
TF-IDF + Logistic Regression	0.690	0.499	0.77	0.66	0.06
mBERT	0.821	0.550	0.87	0.79	0.00
XLM-RoBERTa	0.850	0.571	0.89	0.82	0.00
KLUE-RoBERTa	0.868	0.642	0.91	0.85	0.17

Table1. summarizes the performance of the baseline, multilingual, and Korean-specific models on the consolidated KOTE sentiment classification task.

(a) Model comparison for accuracy and macro F1.



(b) Confusion matrix for KLUE-RoBERTa

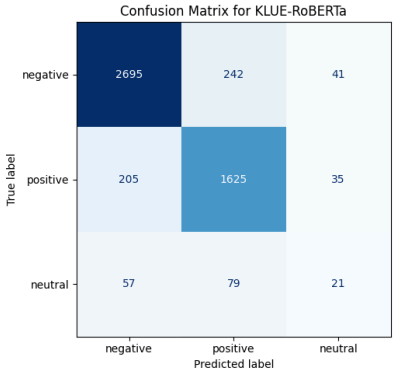


Figure 2. (a) Comparison of all models using accuracy and macro F1; (b) confusion matrix for KLUE-RoBERTa on the test set.

The experiments showed that all transformer-based models performed better than the traditional TF-IDF baseline. The TF-IDF with logistic regression model achieved 69.0% test accuracy and 0.499 macro F1. This shows that a simple bag-of-words approach was able to separate positive and negative sentiment to some extent, but it still had clear limitations, especially for the smaller neutral class.

Among the transformer models, the Korean-specific model performed the best overall. KLUE-RoBERTa achieved 86.8% accuracy and 0.642 macro F1, which was the highest result in this project. XLM-RoBERTa achieved 85.0% accuracy and 0.571 macro F1, while multilingual BERT achieved 82.1% accuracy and 0.550 macro F1. These results suggest that pretrained contextual models are much more effective than TF-IDF features for this task, and that Korean-specific pretraining gives a clear advantage over multilingual pretraining for Korean sentiment classification.

The class-level results also showed an important pattern. All models performed much better on the negative and positive classes than on the neutral class. For KLUE-RoBERTa, the F1 score was 0.91 for negative and 0.85 for positive, but only 0.17 for neutral. Multilingual BERT and XLM-RoBERTa failed to predict any neutral labels, resulting in an F1 score of 0.00 for both models. The TF-IDF model was still able to predict a few neutral examples because it relied on direct words and phrases and was forced to pay attention to the smaller class through class balancing. In contrast, mBERT and XLM-RoBERTa mainly learned the larger negative and positive classes and ended up classifying neutral examples as either negative or positive.

The neutral-class error analysis based on the KLUE-RoBERTa results helps explain this difficulty. Out of 157 true neutral examples, the model predicted 79 as positive and 57 as negative, while only 21 were correctly classified as neutral. Many of these examples still included labels such as 안타까움/실망 (disappointment/pity), 즐거움/신남 (pleasure/excitement), and 불안/걱정 (anxiety/worry), which are not simple neutral emotions. This suggests that the neutral category is semantically heterogeneous. Even labels that were explicitly mapped to neutral, such as 놀람 (surprise) and 깨달음 (realization), did not behave

consistently as neutral in context. This suggests that the difficulty of the neutral class was caused not only by class imbalance, but also by the score-based relabeling rule, which did not fully reflect the real emotional context of the text. Some comments were labeled as neutral after positive and negative scores canceled out, even though the text itself still expressed clear emotional signals.

Overall, the results suggest that consolidating KOTE into broader sentiment categories made the task more practical, but the neutral category remained a limitation. The model comparison still provides a clear conclusion: transformer-based models outperform the TF-IDF baseline, and Korean-specific pretraining is more effective than multilingual pretraining for this Korean sentiment classification task. At the same time, the neutral-class results show that label consolidation choices can strongly affect model behavior and should be interpreted carefully.

5. Conclusion

This project studied Korean sentiment classification using the KOTE dataset after simplifying its original emotion labels into positive, negative, and neutral categories. The results showed that transformer-based models performed much better than the TF-IDF baseline. Among all models, KLUE-RoBERTa achieved the best overall performance, which suggests that Korean-specific pretraining is more effective than multilingual pretraining for this task. At the same time, all models struggled with the neutral class, showing that the relabeling process made that class less clear and harder to predict.

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